Preliminary.

Thm. Let G be a group, and $H, K \leq G$ be subgroups.
Given $x \in HK$, $\#\{(h,k) \in H \times K \mid hk = x\} = \#H \cap K$.

Proof. Define equivalent relation $\sim$ on $H \times K$ as

$$(h_1, k_1) \sim (h_2, k_2) \overset{\text{def}}{\Leftrightarrow} h_1 k_1 = h_2 k_2.$$

Let $[(h,k)] \in H \times K / \sim$ be an arbitrary classes. Define

$$\varphi : [(e,e)] \to [(h,k)] : (a, a^{-1}) \mapsto (ha, a^{-1}k)$$

(Notice that $[(e,e)] = \{(a,b) \in H \times K \mid ab = e\} = \{(a, a^{-1}) \mid a \in H \cap K\}$.
It is well-defined, since $haa^{-1}k = hk$. Moreover, it is bijection,
Injectiveness: given $a_1, a_2 \in H \cap K$,

$$(ha_1, a_1^{-1}k) = (ha_2, a_2^{-1}k) \text{ implies } a_1 = a_2.$$

Surjectiveness: Let $(h', k') \in H \times K$ be given such that $h'k' = hk$.
Then, $h^{-1}h' = kk'^{-1} \in H \cap K$. Put $a = h^{-1}h' = kk'^{-1}$. Then

$$\varphi(a, a^{-1}) = (ha, a^{-1}k) = (h(h^{-1}h'), (k'k^{-1})k) = (h', k'), \text{ so shown.}$$

Now, since $\#[(e,e)] = \#H \cap K$ and

$$\#[(e,e)] = \#[(h,k)] \text{ for any } (h,k) \in H \times K, \text{ so proved.} \quad \square$$

[Semi-Direct Product.
Thm. Let H, K be Groups, and $\varphi: K \to Aut(H)$ be a Homomorphism. Define an action of K on H as

$$\cdot : K \times H \to H : (k, h) \mapsto \varphi(k)(h)$$

Define $G \overset{def}{=} H \times K$ with operation $(h_1, k_1)(h_2, k_2) \overset{def}{=} (h_1 k_1 \cdot h_2, k_1 k_2)$

Then, G is a group under the operation. Moreover, denote

$H \times \{e_K\} := \{(h, e_K) \mid h \in H\} \subseteq G$ and $\{e_H\} \times K := \{(e_H, k) \mid k \in K\}$.

Then, $H \times \{e_K\} \trianglelefteq G$ and $\{e_H\} \times K < G$.

Proof.